Extremal Klein Four Counts in Semi-Extraspecial 2-Groups

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Abstract

Let G be a semi-extraspecial 2-group with $Z(G) = G' = \Phi(G) \cong C_2^2$ and $\dim_{\mathbb{F}_2} G/Z(G) = 2n$. We consider the branch with the largest possible number of elements satisfying $g^2 = 1$, including the identity. We give sharp bounds for the number of actual Klein four subgroups, with separate formulas for odd and even half-dimension and a small exception at $n = 2$. We describe equality through the radical and sign of a sum of three dual quadratic forms. For every even $n \geq 4$, the groups attaining the lower bound constitute one isomorphism type: an eight-dimensional quadratic core together with hyperbolic $\mathbb{F}_4$-planes, all sharing the same two-dimensional target. The corresponding group operation is a central product. The proof retains every $\mathbb{F}_2$-linear refinement and rules out negative scalar-sign compensation between the core and the entire semisimple remainder. Individual planes in an initial splitting of the remainder need not all be positive. Two explicit examples delimit the inference from maximality and the scope of the even-parameter classification.

Keywords: semi-extraspecial groups; Klein four subgroups; quadratic maps; bent functions; central products.

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1 The problem and the main results

Counting elements whose squares are the identity does not determine how those elements commute. The number of Klein four subgroups measures a specific part of this additional structure: it counts pairs of distinct nonidentity involutions whose product is again an involution. For special 2-groups, squares and commutators are encoded by a quadratic map and its polar. Requiring every central quotient of order two to be extraspecial makes all nonzero scalar polars nondegenerate. In the case of a center of order four there are exactly three scalar components, so both the element count and the subgroup count can be expressed by a small, coupled family of quadratic character sums.

We study the branch on which the element count is largest. This is also the totally orthogonal branch in the group class below. Our question is how far the Klein four count can vary on this branch, and which groups attain its lower endpoint. The restriction fixes a natural extremal condition while leaving the interaction of the three square forms to be determined.

All groups are finite. A nonabelian 2-group E is *extraspecial* if $Z(E) = E' = \Phi(E)$ has order 2. A group G is *semi-extraspecial* if G/U is extraspecial for every subgroup U of index 2 in $Z(G)$. Throughout this paper we impose

$$Z(G) = G' = \Phi(G) = W \cong C_2^2, \qquad V = G/W \cong \mathbb{F}_2^{2n}, \qquad |G| = 2^{2n+2}, \tag{1.1}$$

and require G to be semi-extraspecial. The parity parameter is n, the *half-dimension* of V.

Write

$$I(G) = \{g \in G : g^2 = 1\}, \quad s(G) = |I(G)|, \quad v_4(G) = |\{H \leq G : H \cong C_2 \times C_2\}|.$$

Thus $s(G)$ includes the identity and $v_4(G)$ counts subgroups, not their conjugacy classes. Put

$$h = 2^n, \qquad N = h^2, \qquad M_n = \frac{h^4 + 9h^3 + 36h^2 - 36h + 8}{24}. \tag{1.2}$$

Theorem 1.1 (Sharp numerical bounds)**.** *There is no group in* (1.1) *with* $n = 1$*. For each* $n \geq 2$ *the class is nonempty and*

$$s(G) \leq 2^{2n} + 3 \cdot 2^n = N + 3h,$$

with equality attainable. On the branch $s(G) = N + 3h$*, the sharp bounds are*

$$\begin{aligned} &n = 2: && M_n \leq v_4(G) \leq M_n + 2^{3n-2}, \\ &n \geq 3 \text{ odd}: && M_n - 2^{3n-5} \leq v_4(G) \leq M_n + 2^{3n-5}, \\ &n \geq 4 \text{ even}: && M_n - 2^{3n-4} \leq v_4(G) \leq M_n + 2^{3n-2}. \end{aligned} \tag{1.3}$$

Every displayed endpoint is attained for every indicated n*. For* $n = 2$*, the only values on this branch are* 53 *and* 69*. No assertion is made that every integer between the endpoints occurs.*

The square map is the $\mathbb{F}_2$-quadratic map $Q : V \to W$, $Q(gW) = g^2$, with W written additively. Two such maps $Q : V \to W$ and $\widetilde{Q} : \widetilde{V} \to \widetilde{W}$ are *strictly equivalent* if there are invertible $\mathbb{F}_2$-linear maps $S : V \to \widetilde{V}$ and $T : W \to \widetilde{W}$ such that

$$\widetilde{Q}(Sv) = TQ(v) \qquad (v \in V). \tag{1.4}$$

In particular, adding an arbitrary linear map is not part of this equivalence. Its precise group-theoretic meaning is proved in proposition 2.1.

Let $K = \mathbb{F}_4 = \mathbb{F}_2[\alpha]$, where $\alpha^2 + \alpha + 1 = 0$. The trace $\mathrm{Tr} = \mathrm{Tr}_{K/\mathbb{F}_2}$ is regarded as K-valued through $\mathbb{F}_2 \subset K$. Define

$$Q_*(a, b, c, d) = ad + bc + \mathrm{Tr}(ac) + a + \alpha c, \qquad H_+(u, v) = uv. \tag{1.5}$$

Both are $\mathbb{F}_2$-quadratic maps with target K. The use of K-coordinates does *not* declare Q_* to be a homogeneous quadratic form over K.

Theorem 1.2 (Even-parameter lower equality)**.** *Assume* (1.1)*, and let* $n \geq 4$ *be even. The following conditions are equivalent:*

(i) $s(G) = 2^{2n} + 3 \cdot 2^n$ *and* $v_4(G) = M_n - 2^{3n-4}$*.*

(ii) The square map of G *is strictly equivalent to*

$$Q_* \perp H_+^{\perp r}, \qquad r = (n-4)/2,$$

where all summands take values in the same $W = K$*.*

(iii) G *is isomorphic to the central product, along that common* W*, of the group* G_* *associated with* Q_* *and* r *copies of the order-*64 *Heisenberg group* $H(K)$*.*

Consequently there is exactly one group isomorphism type for each fixed even $n \geq 4$ *satisfying (i). This states existence of a standard decomposition and uniqueness of the whole group type, not uniqueness of an embedded core or of a central decomposition.*

Table 1: Sharp Klein four counts on the maximum-s branch.

n	$\lvert G\rvert$	$s_{\max}$	$\min v_4$	$\max v_4$
2	64	28	53	69
3	256	88	431	463
4	1024	304	4371	5651

The first numerical instances of theorem 1.1 are listed in table 1. The first higher-dimensional instance of theorem 1.2 has $n = 6$, $\dim_{\mathbb{F}_2} V = 12$, and $|G| = 16384$. The classification does not include odd n, other values of s, or the intermediate attainable values of v_4.

The methods belong to existing frameworks. Quadratic maps model special 2-groups and their central products (Kaur and Kulshrestha, 2012, 2015); alternating pencils describe the commutator structure (Lewis and Maglione, 2020; Plût et al., 2014; Macario-Rat et al., 2013). The scalar-dual description of (Abdukhalikov et al., 2024, Theorem 2.1, pp. 509–510) and the general Maiorana–McFarland dual formula of (Çeşmelioğlu et al., 2018, equation (4), p. 313) provide the character-sum tools used below. The $D \equiv 0$ upper endpoint is a direct consequence of the partial-difference-set theorem in (Wang et al., 2023, Theorem 6), combined with an elementary subgroup lift count. We give that substitution in section 8.

The results proved here assemble the numerical bounds, their precise equality conditions, and the exhaustive lower-equality classification in theorem 1.2. The selection of the 96 refinements of the equality polar, their local normalization, and the sign restriction between the core and the entire remainder are short deductions from the cited formulas and classical quadratic-form criteria. They are not new general duality or equivalence methods. The distinction from a classification of polar pencils is that the full square map, including its linear terms, must be retained. All the arguments needed for the stated results are included; no computational enumeration is used as a proof of exhaustiveness.

2 Square maps, strict equivalence, and counting

An $\mathbb{F}_2$-quadratic map $Q : V \to W$ is a map with $Q(0) = 0$ for which

$$B(x, y) = Q(x + y) + Q(x) + Q(y)$$

is bilinear. Its polar B is alternating. Linear terms are allowed and carry information.

Proposition 2.1 (The group model)**.** *For the groups under consideration, $Q(gW) = g^2$ is well-defined, its polar is the commutator map, and all three nonzero scalar polars λB, $\lambda \in W^* \setminus \{0\}$, are nondegenerate. Conversely, any $\mathbb{F}_2$-quadratic map from a nonzero V to $W = \mathbb{F}_2^2$ with these three polars nondegenerate is the square map of a group satisfying* (1.1) *and the semi-extraspecial condition. Two resulting groups are isomorphic if and only if their square maps are strictly equivalent in the sense of* (1.4)*.*

Proof. The hypotheses $G' = W \leq Z(G)$, $W^2 = 1$, and G/W elementary abelian imply that G has class 2 and exponent 4. Changing a lift by an element of W changes neither its square nor a commutator. The square identity in a class-2 group gives the polar formula. For $U = \ker \lambda$, the center of G/U modulo W/U is the radical of λB. Thus G/U is extraspecial exactly when this radical is zero.

For the converse, choose coordinates and write

$$Q(v) = \sum_i a_i v_i + \sum_{i<j} c_{ij} v_i v_j.$$

The bilinear map

$$F(v,u)=\sum_i a_i v_i u_i+\sum_{i<j} c_{ij} v_i u_j$$

has $F(v,v)=Q(v)$. Define on $V\times W$

$$(v,w)(u,t)=(v+u,w+t+F(v,u)). \tag{2.1}$$

Bilinearity proves the cocycle identity and associativity. The square and commutator are Q and B. Nondegeneracy of the scalar polars gives $\operatorname{rad} B=0$, so the center is exactly W. The span of the values of B is W, since otherwise a nonzero λ would annihilate B. Hence the derived subgroup is W; the identity $\Phi(G)=G^2G'$ then gives $\Phi(G)=W$. The preceding quotient argument proves the semi-extraspecial condition.

A group isomorphism preserves the characteristic center and squares, and therefore induces S,T in (1.4). Conversely, a basis of V, lifted to G, together with a basis of the elementary abelian center presents G by centrality, its generator squares $Q(e_i)$, and its commutators $B(e_i,e_j)$. Collection puts every element in a form $wg_1^{a_1}\cdots g_{2n}^{a_{2n}}$, $a_i\in\{0,1\}$. The presentation has at most $4\cdot 2^{2n}$ elements and surjects onto G, so these are all its relations. Sending the center by T and each g_i to a lift of Se_i respects these relations by (1.4). It gives a surjection between groups of equal order and hence an isomorphism. □

Lemma 2.2 (A common target and a central product)**.** *If $V=V_0\oplus V_1$ and $B(V_0,V_1)=0$, then $Q=Q|_{V_0}\perp Q|_{V_1}$ as maps to the* same *W. When the restrictions have nondegenerate scalar polars, the corresponding group is their central product along W. For $r+1$ factors this means*

$$(G_0\times\cdots\times G_r)/\mathcal{N},\qquad \mathcal{N}=\{(w_0,\dots,w_r)\in W^{r+1}:\ \textstyle\sum_i w_i=0\}.$$

Proof. The assertion about Q is exactly its polar identity. Inside the group associated with Q, the preimages of V_0,V_1 commute, intersect in W, and generate the group. Multiplication from their external direct product has the stated central kernel. The same argument applies to any number of summands. This also proves the square-map version of the standard correspondence (Kaur and Kulshrestha, 2012, Lemma 2.4). □

Let $S=Q^{-1}(0)$, $z=|S|$, and set

$$T_0=|\{(x,y)\in S^2: x+y\in S\}|,\qquad J_2=|\{(g,h)\in I(G)^2: gh=hg\}|.$$

Lemma 2.3 (Counting actual Klein four subgroups)**.** *One has*

$$s=4z,\qquad J_2=16T_0=3s-2+6v_4(G). \tag{2.2}$$

If L is the number of two-dimensional $\mathbb{F}_2$-subspaces on which Q vanishes, then

$$T_0=3z-2+6L,\qquad v_4(G)=16L+6z-5. \tag{2.3}$$

Proof. Each zero vector has four involutory lifts, independently for each member of an ordered pair. For $x,y\in S$, the lifts commute exactly when $B(x,y)=Q(x+y)=0$. This proves the first two equalities. Among commuting ordered pairs of involutory elements, there are s diagonal pairs, $s-1$ pairs with only the first entry the identity, and $s-1$ with only the second. Every remaining pair generates a unique Klein four group; each such subgroup has six ordered generating pairs. The identical argument in S, with zero in place of the identity, gives $T_0=3z-2+6L$ and then (2.3). □

3 A third Walsh moment and the sharp bounds

Choose a basis of W^*, write the scalar components as q_1, q_2, and put $q_3 = q_1 + q_2$. For $\xi \in V^*$, define

$$\mathcal{W}_i(\xi) = \sum_{x \in V} (-1)^{q_i(x)+\xi(x)}, \qquad a_0 = \sum_{i=1}^{3} \mathcal{W}_i(0), \qquad \mathcal{P} = \sum_{\xi \in V^*} \prod_{i=1}^{3} \mathcal{W}_i(\xi).$$

We use the Fourier convention $\widehat{f}(\xi) = \sum_x f(x)(-1)^{\xi(x)}$.

Lemma 3.1 (Quadratic character sums)**.** *A quadratic function d on a finite $\mathbb{F}_2$-space with $d(0) = 0$ and polar radical of dimension ρ has character sum zero if its restriction to that radical is nonzero. Otherwise, if the ambient dimension is $2n$, its character sum is $\pm 2^{n+\rho/2}$. For a nondegenerate scalar quadratic form q, every Walsh value has absolute value 2^n.*

Proof. On the radical, d is linear. Translating by a radical vector with value 1 pairs terms with opposite signs. If the restriction is zero, d descends to a nondegenerate quadratic form on the quotient, with 2^ρ preimages for each vector. A symplectic basis decomposes a nondegenerate alternating polar into two-dimensional planes. On such a plane the quadratic form is $xy + ax + by$, whose character sum is $2(-1)^{ab}$. Multiplication over the planes proves the formula. Adding a linear form to a nondegenerate quadratic form changes no polar, which proves the Walsh assertion. □

Proposition 3.2 (The counting identity)**.** *The maximum possible value of s is at most $N + 3h$. It is reached if and only if all three original Gauss sums $\mathcal{W}_i(0)$ equal h. Without assuming this equality, one has*

$$T_0 = \frac{N^2 + 3Na_0 + 3a_0^2 + 21N}{64} + \frac{3\mathcal{P}}{32N}. \tag{3.1}$$

On the branch $s = N + 3h$,

$$v_4(G) = M_n + \frac{\mathcal{P}}{4N}. \tag{3.2}$$

Proof. The indicator identity

$$1_S = \frac{1 + (-1)^{q_1} + (-1)^{q_2} + (-1)^{q_3}}{4}$$

gives $z = (N + a_0)/4$ and $s = N + a_0$. By lemma 3.1, each original Gauss sum is $\pm h$, proving the maximum assertion and its equality criterion.

Put $U(\xi) = \sum_i \mathcal{W}_i(\xi)$. Then $\widehat{1_S}(\xi) = (N\mathbf{1}_{\xi=0} + U(\xi))/4$. Fourier orthogonality gives $T_0 = N^{-1} \sum_\xi \widehat{1_S}(\xi)^3$. Since $\mathcal{W}_i(\xi)^2 = N$,

$$U(\xi)^3 = 7NU(\xi) + 6\mathcal{W}_1(\xi)\mathcal{W}_2(\xi)\mathcal{W}_3(\xi).$$

Moreover $\sum_\xi U(\xi) = 3N$, because each $q_i(0) = 0$. Thus the numerator after expansion is $N^3 + 3N^2a_0 + 3Na_0^2 + 21N^2 + 6\mathcal{P}$, divided by $64N$. This proves (3.1). Substituting $a_0 = 3h$ and (2.2), using $N = h^2$, gives (3.2). □

For a nondegenerate binary quadratic form, its actual Walsh dual records the hyperbolic or elliptic type of every linear refinement; this is (Abdukhalikov et al., 2024, Theorem 2.1, pp. 509–510). In binary coordinates, a diagonal square represents a linear function, so the statement includes the refinements used here. Let B_i be the polar matrices of q_i in the

chosen binary coordinates. More explicitly, if $\mathcal{W}_i(0) = h(-1)^{\epsilon_i}$ and covectors are columns, translation by $B_i^{-1}\xi$ gives

$$\mathcal{W}_i(\xi) = h(-1)^{q_i(B_i^{-1}\xi)+\epsilon_i}.$$

The origin phase ϵ_i is part of the actual dual. We now assume $s = N + 3h$, so all three ϵ_i vanish. Completing the square gives

$$\mathcal{W}_i(\xi) = h(-1)^{q_i^*(\xi)}, \qquad q_i^*(\xi) = q_i(B_i^{-1}\xi), \qquad q_i^*(0) = 0. \tag{3.3}$$

Indeed translation by $B_i^{-1}\xi$ removes the Fourier linear term. Define

$$D = q_1^* + q_2^* + q_3^*, \qquad C = B_1^{-1} + B_2^{-1} + (B_1 + B_2)^{-1}, \qquad \rho = \dim\ker C. \tag{3.4}$$

Here D is the pointwise sum in $\mathbb{F}_2$, whereas $\mathcal{G}(D) = \sum_\xi (-1)^{D(\xi)}$ is an integer. Thus $D \equiv 0$ gives $\mathcal{G}(D) = N$, not zero. The matrix C is the polar of D. In particular

$$\mathcal{P} = h^3 \sum_\xi (-1)^{D(\xi)}. \tag{3.5}$$

Each inverse B_i^{-1} is alternating: for $\xi = B_i x$, $\xi^\mathsf{T} B_i^{-1}\xi = x^\mathsf{T} B_i x = 0$.

Lemma 3.3 (Paired cyclic summands)**.** *Let β be a nondegenerate alternating form over $\mathbb{F}_2$, and let A be self-adjoint for β, with $\beta(v, Av) = 0$ for every v. There is a β-orthogonal decomposition into A-invariant nondegenerate spaces $X \oplus Y$, where X, Y are totally isotropic cyclic modules with the same elementary divisor f^k. In particular every elementary divisor occurs in pairs.*

Proof. Self-adjointness gives

$$\beta(v, A^{2j}v) = \beta(A^j v, A^j v) = 0,$$
$$\beta(v, A^{2j+1}v) = \beta(A^j v, A^{j+1} v) = 0.$$

Every cyclic module is therefore totally isotropic. Distinct primary components are orthogonal: on one of them a polynomial annihilating the other is invertible, and self-adjointness moves that polynomial to the other argument. Hence each primary component is nondegenerate.

In an f-primary component choose v with maximal annihilator f^k and let $X = \mathbb{F}_2[A]v$. Choose w in that component with $\beta(f(A)^{k-1}v, w) \neq 0$, and put $Y = \mathbb{F}_2[A]w$. The kernel of the pairing map $X \to Y^*$ is a submodule. Any nonzero submodule of $\mathbb{F}_2[t]/(f^k)$ contains its socle $f^{k-1}X$, but the chosen pairing is not zero on that socle. The kernel is zero. Maximality of k gives $\dim Y \leq \dim X$; thus the pairing is perfect and dimensions are equal. Total isotropy and perfect pairing imply $X \cap Y = 0$ and nondegeneracy of $X \oplus Y$. Its orthogonal complement is A-invariant. Repeating proves the lemma. □

Proposition 3.4 (Radical restrictions)**.** *Set $A = B_1^{-1}B_2$ and $p(t) = t^2 + t + 1$. Then $A, I + A$ are invertible and*

$$C = [A(I + A)]^{-1} p(A) B_1^{-1}, \qquad \ker C = B_1 \ker p(A). \tag{3.6}$$

Consequently ρ is divisible by 4. If n is odd, then $\rho \leq 2n-6$. If n is even and $\sum_\xi (-1)^{D(\xi)} < 0$, then $\rho \leq 2n - 4$.

Proof. Writing $B_2 = B_1 A$ and $B_3 = B_1(I + A)$ proves invertibility and (3.6) by direct multiplication. Notice that the factor B_1 in the kernel equality must be retained. For $\beta(x, y) = x^\mathsf{T} B_1 y$, the matrix A is self-adjoint, and $\beta(v, Av) = 0$, because B_2 is alternating. Apply lemma 3.3. A pair of p^k-blocks occupies $4k$ dimensions and contributes 4 to $\dim\ker p(A)$.

A pair belonging to any other irreducible polynomial contributes zero to this kernel. There are no linear factors, and p is the only irreducible quadratic over $\mathbb{F}_2$; every other pair therefore occupies at least six dimensions.

If n is odd, the identity $n = \sum_{\text{pairs}} k \deg f$ forces at least one pair with both k and $\deg f$ odd. Its degree is at least three, so the non-p part occupies at least six dimensions. This gives $\rho \leq 2n - 6$. For even n, if $\rho = 2n$, then $C = 0$ and D is linear with $D(0) = 0$. Its sum is N or 0, never negative. Divisibility by four then yields $\rho \leq 2n - 4$ in the negative case. □

Proposition 3.5 (Bounds and exact equality tests)**.** *On the maximum-s branch, the inequalities in* (1.3) *hold. Their equality tests are:*

(a) For odd $n \geq 3$, either endpoint requires and is implied by: one pair of simple irreducible cubic blocks and all remaining pairs simple p-blocks; $D|_{\ker C} = 0$; and the Gauss sum of the six-dimensional quotient is respectively -8 or $+8$.

(b) For even $n \geq 4$, lower equality requires and is implied by: one pair of p^2-blocks and all remaining pairs simple p-blocks; $D|_{\ker C} = 0$; and the Gauss sum of the four-dimensional quotient is -4.

(c) For even $n \geq 2$, upper equality holds if and only if $D = 0$. Equivalently $p(A) = 0$ and the remaining linear function D vanishes identically.

(d) For $n = 2$, $C = 0$; the lower value 53 holds if and only if D is a nonzero linear function, and the upper value 69 if and only if $D = 0$.

Proof. By lemma 3.1, the sum for D is zero or $\pm 2^{n+\rho/2}$, with the nonzero case exactly when D vanishes on its polar radical. Combining proposition 3.4 with (3.5) gives $|\mathcal{P}| \leq 2^{5n-3}$ for odd n. For even n, the positive bound is $\mathcal{P} \leq 2^{5n}$; a negative value satisfies $\mathcal{P} \geq -2^{5n-2}$. Equation (3.2) gives the inequalities.

For the odd endpoints, equality in magnitude forces $\rho = 2n - 6$. The deficit $2n - \rho$ contributed by a p^k-pair is $4(k - 1)$; a non-p pair contributes $2k \deg f \geq 6$. The odd-parity argument forces such a non-p pair, and deficit six therefore means precisely one simple cubic pair and otherwise simple p-pairs. The remaining equality conditions follow from the quadratic sum on the quotient. Conversely these conditions give exactly the asserted magnitude and sign. Deficit four similarly forces exactly one p^2-pair, proving (b) in both directions. To attain the even upper bound, every term $(-1)^{D(\xi)}$ must be $+1$, so $D = 0$; this also forces $p(A) = 0$ by (3.6). In dimension four there is only one simple p-pair, hence $C = 0$; the two possible linear sums give (d). In dimension two the space of alternating forms has dimension one, so three nonzero scalar polars cannot all be nondegenerate. □

4 Attainment of all numerical endpoints

The following constructions prove the sharpness assertions without using the classification theorem. The trace-product positive blocks are standard Maiorana–McFarland constructions (Çeşmelioğlu et al., 2018, Theorem 2(i), p. 314); the calculations retain the parameter correspondence and the signs needed here.

Lemma 4.1 (The four-dimensional blocks)**.** *Identify W with $K = \mathbb{F}_4$ by*

$$\iota(w) = (\mathrm{Tr}(w), \mathrm{Tr}(\alpha w)). \tag{4.1}$$

The map $H_+(u, v) = uv$ has all scalar Gauss sums $+4$, dual function $D \equiv 0$, and $\mathcal{P} = 2^{10}$. The map $uv + u$ has the same polar and the same $s = 28$, but its Klein four count is 53, whereas that of H_+ is 69.

Proof. For $z \neq 0$,

$$\sum_{u,v \in K} (-1)^{\mathrm{Tr}(zuv+\rho u+\sigma v)} = 4(-1)^{\mathrm{Tr}(\rho\sigma/z)}.$$

The sum of z^{-1} over $K^\times$ is zero, so the dual function vanishes identically and its integer character sum is 16. For $uv + u = u(v + 1)$, the zero set consists of the four vectors $(0, v)$ and the three vectors $(u, 1)$ with $u \neq 0$. Its only totally zero two-dimensional $\mathbb{F}_2$-subspace is the line $u = 0$: two distinct nonzero u's in the second set have a sum outside the zero set, and adding a nonzero vector from the line $u = 0$ also leaves the zero set. For uv, the zero set is the union of the two coordinate lines, which are its only totally zero two-dimensional subspaces. In each case $z = 7$, and (2.3) gives 53 and 69. □

Lemma 4.2 (The six-dimensional endpoint blocks)**.** *Let $L = \mathbb{F}_8 = \mathbb{F}_2[t]/(t^3 + t + 1)$, with absolute trace Tr_8. On L^2 put*

$$q_1(x, y) = \mathrm{Tr}_8(xy), \qquad q_2(x, y) = \mathrm{Tr}_8(txy + cx + dy).$$

For $c = d = 0$ and for $c = 1$, $d = t^2 + t + 1$, the three scalar polars are nondegenerate and all three original Gauss sums are $+8$. The respective dual sums have character sums $+8$ and -8; therefore $\mathcal{P} = +2^{12}$ and -2^{12}.

Proof. The three polar coefficients are $1, t, 1 + t$, all nonzero. For $a \neq 0$, summing first over y gives

$$\sum_{x,y \in L} (-1)^{\mathrm{Tr}_8(axy+cx+dy+ux+vy)} = 8(-1)^{\mathrm{Tr}_8((u+c)(v+d)/a)}. \tag{4.2}$$

When $c = d = 0$, the original sums are $+8$. Let $k = t^{-1} + (1 + t)^{-1} = t + 1$. The dual sum is then $\mathrm{Tr}_8(tuv)$, of character sum $+8$. For the second choice, $d = t^5$, $d/t = t^4 = t^2 + t$, and $d/(1 + t) = t^2$; both traces are zero, so the original signs again are positive. The complete dual sum is

$$D(u, v) = \mathrm{Tr}_8(tuv + kdu + kcv + kcd).$$

Using (4.2), its sum is

$$8(-1)^{\mathrm{Tr}_8(cd(k+k^2/t))} = 8(-1)^{\mathrm{Tr}_8(cd\,k/t)} = -8,$$

because $k + k^2/t = k/t$, $(k/t)d = 1$, and $\mathrm{Tr}_8(1) = 1$. This also shows that the polar of this D is nondegenerate. □

Lemma 4.3 (An eight-dimensional lower block)**.** *There exists a square map on $\mathbb{F}_2^8$ with all original scalar Gauss sums $+16$ and with dual character sum -64.*

Proof. Let U be multiplication by t on $\mathbb{F}_2[t]/((t^2 + t + 1)^2)$, in its polynomial basis, and put

$$E = p(U), \qquad K_0 = U^{-1} + (I + U)^{-1}.$$

Here $E^2 = 0$, $\operatorname{rank} E = 2$, and $I + K_0 = E$, since $U(I + U) = I + E$. Indeed multiplication by p on $\mathbb{F}_2[t]/(p^2)$ has image and kernel equal to the ideal (p), each of binary dimension two. Choose $0 \neq c \in \operatorname{im} E$, z with $z^{\mathsf{T}}c = 1$, and $d = E^{\mathsf{T}}z$. On $\mathbb{F}_2^4 \oplus \mathbb{F}_2^4$, take

$$q_1(x, y) = x^{\mathsf{T}}y, \qquad q_2(x, y) = x^{\mathsf{T}}Uy + c^{\mathsf{T}}x + d^{\mathsf{T}}y.$$

The matrices $I, U, I + U$ are invertible, proving nondegeneracy of the three scalar polars. The two possibly nontrivial original Gauss signs are $d^{\mathsf{T}}U^{-1}c$ and $d^{\mathsf{T}}(I + U)^{-1}c$; both vanish because $Ec = 0$ and E commutes with these matrices. Thus all sums are $+16$.

The same summation used in (4.2), now with ordinary binary dot products, gives the complete dual sum at covectors (a, b) as

$$D(a,b) = b^{\mathsf{T}}(Ea + c) + z^{\mathsf{T}}Ea.$$

Here $K_0 c = c$, $EK_0 = E$, and $Ec = 0$ account for the linear and constant terms. Summing over b leaves precisely the four solutions of $Ea = c$. For each, $z^{\mathsf{T}}Ea = 1$. The sum is therefore $-16 \cdot 4 = -64$. □

Completion of the proof of theorem 1.1. For a common-target orthogonal sum, each scalar Walsh transform is a product of the corresponding transforms. The third moment $\mathcal{P}$ and each original Gauss sum consequently multiply. For even n, $n/2$ copies of H_+ have original sums h and $\mathcal{P} = 2^{5n}$, giving the upper endpoint. Equivalently, in the notation of (Çeşmelioğlu et al., 2018, Theorem 2(i)), take $p = 2$, $s = 2$, $m = n$, $E = \mathbb{F}_{2^n}$, and $\pi(y) = y$, with $g = 0$. Then $\mathrm{Tr}_{E/K}(xy)$ becomes $H_+^{\perp n/2}$ in a pair of trace-dual K-bases. The condition $2 \mid n$ is essential to this subfield description. For odd n, one of the blocks in lemma 4.2 together with $(n-3)/2$ copies of H_+, all with the same ordered target, has $\mathcal{P} = \pm 2^{5n-3}$. For even $n \geq 4$, the block in lemma 4.3 with $(n-4)/2$ copies of H_+ has $\mathcal{P} = -2^{5n-2}$. All original Gauss sums in these constructions are $+h$. Every scalar polar remains nondegenerate, so proposition 2.1 realizes actual groups in the asserted class. Finally lemma 4.1 supplies both $n = 2$ endpoints. Together with proposition 3.5, this proves every assertion of theorem 1.1. □

5 The eight-dimensional square maps

In this section $p(t) = t^2 + t + 1$. We first normalize the ordered polar pair, keeping the target coordinates fixed.

Lemma 5.1 (A simultaneous polar model)**.** *Suppose $A = B_1^{-1}B_2$ has one pair of p^2-blocks and r pairs of p-blocks. There is a common orthogonal decomposition*

$$V = V_0 \oplus V_1, \qquad \dim_{\mathbb{F}_2} V_0 = 8, \quad \dim_{\mathbb{F}_2} V_1 = 4r, \quad B(V_0, V_1) = 0.$$

Put $R = \mathbb{F}_2[t]/(p^2)$, $\varepsilon = p(t)$, and $\alpha = t + \varepsilon = t^2 + 1$. Then $R = K[\varepsilon]/(\varepsilon^2)$, $K = \mathbb{F}_4$. After input changes and the single common target identification (4.1)*, the restrictions of Q have the forms*

$$Q^{(0)} = ad + bc + \mathrm{Tr}(ac) + fa + ga^2 + \ell c + mc^2 + \eta b + kb^2 + \theta d + jd^2, \tag{5.1}$$

$$Q^{(1)} = \sum_{\nu=1}^{r} (u_\nu v_\nu + F_\nu u_\nu + G_\nu u_\nu^2 + L_\nu v_\nu + M_\nu v_\nu^2), \tag{5.2}$$

where all coefficients lie in K. These include all $\mathbb{F}_2$-linear refinements of the fixed polars. Each nonzero scalar polar of each nonempty restriction is nondegenerate.

Proof. Use lemma 3.3 to select the paired p^2-cyclic modules as V_0. Their $\beta = B_1$-orthogonal complement V_1 is A-invariant by self-adjointness. Hence the decomposition is orthogonal for both B_1 and B_2, and thus for B. Each scalar polar is a nondegenerate block diagonal form. The polar identity already gives $Q = Q^{(0)} \perp Q^{(1)}$; it does not yet imply any sign condition on the restrictions.

Let $\lambda : R \to \mathbb{F}_2$ extract the coefficient of t^3. The pairing $\lambda(xy)$ is nondegenerate: if a polynomial $x \neq 0$ has degree $j \leq 3$, multiplying by t^{3-j} gives coefficient 1 at degree three without a reduction. On a paired cyclic summand R^2, the balanced pairing for B_1 has the form $\lambda'(xy' + x'y)$. Nondegeneracy writes $\lambda'(z) = \lambda(ez)$ for a unit $e \in R$; otherwise

multiplication by e would have a kernel. Rescaling one input generator by e^{-1} normalizes the pair to

$$B((x,y),(x',y')) = \big(\lambda(xy'+x'y), \lambda(t(xy'+x'y))\big). \tag{5.3}$$

This uses only input changes. On each simple p-pair the same argument, using the non-degenerate trace pairing on K, gives $(\mathrm{Tr}(uv'+u'v), \mathrm{Tr}(\alpha(uv'+u'v)))$, in the same ordered target.

The relations $\varepsilon^2 = 0$, $\alpha^2+\alpha+1=0$ give the asserted description of R. Direct evaluation on the basis $1, \alpha, \varepsilon, \alpha\varepsilon$ gives

$$\lambda(a+\varepsilon b) = \mathrm{Tr}(b), \qquad \lambda(t(a+\varepsilon b)) = \mathrm{Tr}(a+\alpha b).$$

For $x = a+\varepsilon b$, $y = c+\varepsilon d$, the base refinement $(\lambda(xy), \lambda(txy))$ therefore becomes $ad+bc+\mathrm{Tr}(ac)$ under ι^{-1}. On a simple pair the base refinement becomes uv.

Two $\mathbb{F}_2$-quadratic maps with the same polar differ by an $\mathbb{F}_2$-linear map. Every such map $K \to K$ is uniquely $x \mapsto ux + vx^2$: the map from the two coefficients is injective by evaluating at $1, \alpha$, and both spaces have $\mathbb{F}_2$-dimension four. Applying this independently to all coordinates proves (5.1) and (5.2). No assumption of K-homogeneity has been made. □

The next formula is the specialization of the general Maiorana–McFarland formula in (Çeşmelioğlu et al., 2018, equation (4), p. 313). The common $\mathbb{F}_{16}$ coordinates and the arbitrary function g in that substitution are recorded in appendix A. We also give a direct sum calculation to fix the Fourier convention and the constants.

Lemma 5.2 (The complete core phases)**.** *For* (5.1) *and* $z \in K^\times$*, let* $\mathcal{W}_z$ *be the Walsh transform of* $\mathrm{Tr}(zQ^{(0)})$*. Use the Fourier functional* $\mathrm{Tr}(\rho a + \sigma b + \tau c + \upsilon d)$*, and write* $\mathcal{W}_z = 16(-1)^{E_z}$*. With* $\kappa_z = \mathrm{Tr}(z)$*, one has*

$$a_z = \upsilon/z + \theta + zj^2, \qquad c_z = \sigma/z + \eta + zk^2, \tag{5.4}$$

$$E_z = \mathrm{Tr}\big(\kappa_z a_z c_z + (\rho + zf + z^2g^2)a_z + (\tau + z\ell + z^2m^2)c_z\big). \tag{5.5}$$

Phase additions, including $\sum E_z$ *and* γ *below, are in* $\mathbb{F}_2$*; sums of* $(-1)^{E_z}$ *are integer character sums. In particular the* unadjusted *sum of these three phases is*

$$\sum_{z\neq 0} E_z = \mathrm{Tr}(\sigma\upsilon + (f+\eta)\upsilon + (\ell+\theta)\sigma + \theta\rho + \eta\tau) + \gamma, \tag{5.6}$$

$$\gamma = \mathrm{Tr}(\theta k^2 + \eta j^2 + jk + gj + mk) = \sum_{z\neq 0} E_z(0). \tag{5.7}$$

These formulas hold without a hypothesis on the original Gauss signs.

Proof. The identity $\mathrm{Tr}(za^2) = \mathrm{Tr}(z^2a)$ over K moves each squared-variable linear term into the ordinary Fourier coefficient. Summing first over b, d forces exactly the two equations in (5.4) and leaves a factor $4^2 = 16$. The remaining exponent is (5.5). For the sum over $z \in K^\times$, use

$$\sum z = \sum z^2 = \sum z^{-1} = 0, \quad \sum z^3 = 1, \quad \sum \kappa_z = 0,$$

and

$$\sum \kappa_z/z = \sum \kappa_z/z^2 = \sum \kappa_z z = \sum \kappa_z z^2 = 1.$$

Expansion gives the nonconstant terms in (5.6). The constant terms are $\mathrm{Tr}(\theta k^2 + \eta j^2 + j^2k^2 + g^2j^2 + m^2k^2)$, which is (5.7) since the trace is invariant under squaring. This proves the formula including its constant. □

Lemma 5.3 (The anisotropic sign obstruction)**.** *Suppose $\eta = \theta = 0$ in* (5.1)*, and all three original Gauss sums have the same sign $(-1)^\delta$. Then*

$$fj^2 + \ell k^2 = jk, \qquad \mathrm{Tr}(gj + mk + jk) = \delta. \tag{5.8}$$

If additionally $\mathrm{Tr}(f\ell) = 1$, then $j = k = 0$ and $\delta = 0$.

Proof. At the Fourier origin, $a_z = zj^2$ and $c_z = zk^2$. Put $e = jk,\ u = fj^2 + \ell k^2,\ v = gj + mk$. Using $\mathrm{Tr}(z)\,\mathrm{Tr}(ze) = \mathrm{Tr}(z^2 e) + \mathrm{Tr}(e)$ for $z \neq 0$, formula (5.5) becomes

$$E_z(0) = \mathrm{Tr}(z^2(u + e)) + \mathrm{Tr}(v + e).$$

The three values equal δ precisely when $u + e = 0$ and $\mathrm{Tr}(v + e) = \delta$: a nonzero linear functional on K is not constant on $K^\times$. This proves (5.8).

If $\mathrm{Tr}(f\ell) = 1$, then $f \neq 0$, and the binary K-quadratic form $fX^2 + XY + \ell Y^2$ is anisotropic. Indeed a nonzero solution with $Y \neq 0$ would give, on setting $w = fX/Y$, the equation $w^2 + w + f\ell = 0$, contradicted by its trace. If $Y = 0$, then $X = 0$. The first equation in (5.8) therefore forces $j = k = 0$; the second now forces $\delta = 0$. □

With the equality polar fixed, the following selection is a short application of the preceding dual formula, the radical criterion in lemma 3.1, and the binary Arf test in lemma 5.3.

Proposition 5.4 (The 96 core refinements)**.** *In dimension eight, the maps with maximum $s = 304$ and $v_4 = 4371$, in the fixed polar coordinates* (5.1)*, are exactly*

$$Q_{f,g,\ell,m} = ad + bc + \mathrm{Tr}(ac) + fa + ga^2 + \ell c + mc^2, \qquad \mathrm{Tr}(f\ell) = 1. \tag{5.9}$$

There are 96 *such coordinate maps. Every dimension-eight lower-equality map can be put in these coordinates.*

Proof. By proposition 3.5(b), every lower-equality polar has exactly one pair of p^2-blocks. Thus lemma 5.1 covers all of them, rather than just a selected construction. Maximum s means $E_z(0) = 0$ for all $z \neq 0$, hence $\gamma = 0$. The radical of the polar in (5.6) consists of the directions ρ, τ, with $\sigma = \upsilon = 0$. The radical-vanishing condition therefore forces $\eta = \theta = 0$. Summing over ρ, τ and then completing the square in σ, υ gives

$$\sum_{\rho,\sigma,\tau,\upsilon} (-1)^D = 64(-1)^{\mathrm{Tr}(f\ell)}. \tag{5.10}$$

Lower equality requires -64, hence $\mathrm{Tr}(f\ell) = 1$. Apply lemma 5.3 with $\delta = 0$ to get $j = k = 0$. This proves necessity of (5.9). Conversely, for (5.9), formula (5.5) at the origin gives all three signs positive, and (5.10) is -64. The counting identity gives $v_4 = M_4 - 2^8 = 4627 - 256 = 4371$. There are three choices for $f \neq 0$, two for $f\ell$ with trace one, and four each for g, m, giving $3 \cdot 2 \cdot 4^2 = 96$. □

The local input changes below are instances of the transformation framework of (Macario-Rat et al., 2013, Propositions 5–6, pp. 12–14) and (Plût et al., 2014, Propositions 4.5 and 4.7, p. 17). The last distinction is the classical binary Arf class over K. We retain the explicit changes, including their induced linear terms.

Proposition 5.5 (Explicit transitivity)**.** *All* 96 *maps in* (5.9) *are strictly equivalent to Q_*, using only input transformations that preserve both scalar polars and leave the target fixed.*

Proof. Every matrix in $\mathrm{SL}_2(R)$, acting on $(x, y) = (a + \varepsilon b, c + \varepsilon d)$, preserves $xy' + x'y$ and therefore both forms in (5.3). Starting with (5.9), make the following pullback changes.

1. The matrix
$$M_1 = \begin{pmatrix} 1 & \varepsilon m \\ \varepsilon g & 1 \end{pmatrix}$$
has determinant one and changes b to $b + mc$, d to $d + ga$. Substitution cancels both ga^2 and mc^2.

2. The matrix $M_2 = \mathrm{diag}(f^{-1}, f)$ changes the two remaining coefficients to $1, f\ell$. The latter is α or α^2.

3. If $f\ell = \alpha^2$, use
$$M_3 = \begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix}.$$
The new coefficient of c is $\alpha^2 + \alpha + \alpha^2 = \alpha$, and an extra αc^2 is introduced. The matrix $M_4 = \begin{pmatrix} 1 & \varepsilon\alpha \\ 0 & 1 \end{pmatrix}$ cancels this last term.

The resulting map is exactly (1.5). All the operations act on the entire quadratic map. In particular the linear correction generated by pulling back the base map $ad + bc + \mathrm{Tr}(ac)$ has been included, not discarded. Each operation has target transformation $T = \mathrm{id}_W$. □

Remark 5.6. In fact the full componentwise isometry group of (5.3) is $\mathrm{SL}_2(R)$. Such an isometry commutes with A, hence is R-linear. Its determinant d satisfies $\lambda((d-1)r) = 0$ for every $r \in R$, forcing $d = 1$. There are $16^2 - 4^2 = 240$ possible unimodular first columns, and 16 second columns with determinant one for each: a unimodular column has a unit coordinate, so its determinant map $R^2 \to R$ is onto and each fibre has size 16. Thus its order is 3840. This observation is not needed to infer transitivity from the explicit changes above. Allowing additional target changes cannot split an already single orbit. Thus no uncomputed subgroup orbit is being mistaken for a full classification.

For a concrete group G_*, take $K^4 \times K$ and use (2.1) with
$$F((a,b,c,d),(a',b',c',d')) = ad' + bc' + \mathrm{Tr}(ac') + (aa')^2 + \alpha(cc')^2. \tag{5.11}$$
This is $\mathbb{F}_2$-bilinear, and its diagonal is Q_*. The scalar polars in (5.3) are nondegenerate because $1, t, 1+t$ are units in R. Consequently proposition 2.1 and propositions 5.4 and 5.5 prove theorem 1.2 for $n = 4$, including existence of the group of order 1024.

6 The common core in every even half-dimension

For the remainder of this section let $n = 4 + 2r$, $r \geq 0$. Use the common-target coordinates of lemma 5.1.

Lemma 6.1 (The semisimple phases)**.** *For* (5.2)*, the scalar Walsh transform at the functional* $\mathrm{Tr}\sum_\nu(\rho_\nu u_\nu + \sigma_\nu v_\nu)$ *is* $4^r(-1)^{E_{1,z}}$*, where*
$$E_{1,z} = \sum_\nu \mathrm{Tr}\left(\frac{(\rho_\nu + zF_\nu + z^2G_\nu^2)(\sigma_\nu + zL_\nu + z^2M_\nu^2)}{z}\right). \tag{6.1}$$
Its phase sum is
$$D_1 = \sum_{z\neq 0} E_{1,z} = \mathrm{Tr}\sum_\nu (L_\nu\rho_\nu + F_\nu\sigma_\nu + G_\nu M_\nu). \tag{6.2}$$
If $F_\nu = L_\nu = 0$ *for all* ν*, then each original Gauss sum has the same sign* $(-1)^\delta$*, where* $\delta = \mathrm{Tr}\sum_\nu G_\nu M_\nu$*, and* $D_1 \equiv \delta$*.*

Proof. This is also (Çeşmelioğlu et al., 2018, equation (4), p. 313): on each plane use $p = 2$, $m = 2$, $s = 1$, scalar parameter 1, the pairing $\mathrm{Tr}(\rho u + \sigma v)$, and

$$\pi_z(v) = zv + zF + z^2G^2, \qquad g_z(v) = (zL + z^2M^2)v.$$

The permutation π_z is affine; the original squared-variable terms have been moved by the trace identity, not omitted. Substitution of $\pi_z^{-1}(\rho)$ in that formula gives (6.1), with its origin phase intact. Summing its expansion over $K^\times$, the terms with z, z^2, z^{-1} vanish and $z^3 = 1$; trace invariance under squaring gives (6.2). With $F_\nu = L_\nu = 0$, evaluating (6.1) at the origin gives $\mathrm{Tr}\sum_\nu G_\nu M_\nu = \delta$ for each z. □

The lower-equality sign is essential in the following deduction from the two phase formulas. Positivity of the whole map alone does not give positivity on each restriction.

Lemma 6.2 (Exclusion of core–remainder negative-sign compensation)**.** *Suppose $Q = Q^{(0)} \perp Q^{(1)}$ is in the form* (5.1)–(5.2)*, all three original Gauss sums are $+2^n$, and*

$$D|_{\ker C} = 0, \qquad \sum_{\xi \in V^*} (-1)^{D(\xi)} = -2^{2n-2}. \tag{6.3}$$

Then

$$\eta = \theta = j = k = 0, \quad \mathrm{Tr}(f\ell) = 1, \quad F_\nu = L_\nu = 0 \ (\textit{all } \nu), \quad \mathrm{Tr}\sum_\nu G_\nu M_\nu = 0.$$

In particular both restrictions separately have all three original Gauss sums positive.

Proof. Write the *unadjusted* phases on the two blocks as $E_{0,z}$ and $E_{1,z}$. The product rule for Walsh transforms and global maximality imply only

$$E_{0,z}(0) = E_{1,z}(0) \qquad (z \neq 0). \tag{6.4}$$

The two sides are not yet known to be zero or to be independent of z. The global normalized dual sum is $D = D_0 + D_1$, where $D_j = \sum_{z \neq 0} E_{j,z}$. It has $D(0) = 0$, although its block summands may have constants.

The polar of D_1 is zero by (6.2), so the whole space V_1^* is in the radical of the global polar C. The equation $D(0, \xi_1) = 0$ for every $\xi_1 \in V_1^*$ makes D_1 constant. Trace nondegeneracy in (6.2) forces $F_\nu = L_\nu = 0$ for all ν. By lemma 6.1, put $\delta = \mathrm{Tr}\sum_\nu G_\nu M_\nu$; then $D_1 \equiv \delta$ and $E_{1,z}(0) = \delta$ for every z. Equation (6.4) now gives $E_{0,z}(0) = \delta$, and hence $\gamma = \delta$ in (5.6).

The remaining radical directions are the core coordinates ρ, τ. On them the global D is $\mathrm{Tr}(\theta\rho + \eta\tau) + \delta + \delta$. Radical vanishing forces $\eta = \theta = 0$. The constants cancel between the blocks; they have not been individually suppressed. The complete phase is now

$$D = \mathrm{Tr}(\sigma\upsilon + f\upsilon + \ell\sigma),$$

independent of V_1^*, ρ, and τ. Its character sum is

$$2^{4r}\, 64(-1)^{\mathrm{Tr}(f\ell)}.$$

The *negative* sign in (6.3) therefore forces $\mathrm{Tr}(f\ell) = 1$. The original core signs all equal $(-1)^\delta$, so lemma 5.3 forces $j = k = 0$ and $\delta = 0$. This proves every assertion. Empty sums give $\delta = 0$ when $r = 0$. □

Lemma 6.3 (The positive remainder)**.** *A map*

$$q(u, v) = \sum_{\nu=1}^{r} (u_\nu v_\nu + G_\nu u_\nu^2 + M_\nu v_\nu^2), \qquad \mathrm{Tr}\sum_\nu G_\nu M_\nu = 0, \tag{6.5}$$

is isometric over K, with target unchanged, to $H_+^{\perp r}$.

Proof. At this stage, unlike (5.2), the map really is a homogeneous quadratic form over K. The hyperbolic/elliptic classification recalled in (Abdukhalikov et al., 2024, p. 509) now applies over K; equivalently its component signs can be read from (Abdukhalikov et al., 2024, Theorem 2.2, pp. 511–512) with a trace-compatible pairing. For $r \geq 1$, identify K^{2r} linearly with $\mathbb{F}_{4^{2r}}$: the theorem's base field is K and its extension degree is $2r$. The required K-homogeneity has just been established, rather than inferred from the notation in lemma 5.1. Its polar is the nondegenerate K-symplectic form $\sum_\nu (u_\nu v'_\nu + u'_\nu v_\nu)$. By lemma 6.1, all three scalar Gauss sums are $+4^r$, and its zero count is

$$\frac{4^{2r} + 3 \cdot 4^r}{4}.$$

For $r \geq 1$ this exceeds one, so choose a nonzero isotropic vector u. Choose v with $B_K(u, v) = 1$ and replace it by $v + q(v)u$. Homogeneity and $q(u) = 0$ show that the new vector is isotropic and still pairs to one with u. They span a hyperbolic K-plane. Its K-orthogonal complement has nondegenerate polar and all scalar Gauss sums $+4^{r-1}$, by dividing out the three $+4$ factors. Repetition yields r hyperbolic planes. The resulting input change is K-linear symplectic and fixes W. The case $r = 0$ is the empty space. □

Proof of theorem 1.2. Assume (i). By proposition 3.5(b), the polar has one pair of p^2-blocks and $r = (n-4)/2$ simple p-pairs, the global D vanishes on its radical, and its character sum is -2^{2n-2}. The common model from lemma 5.1 applies. By lemma 6.2, the core is in (5.9) and the remainder is in (6.5). Normalize the core by proposition 5.5 and the remainder by lemma 6.3. Both input changes have $T = \mathrm{id}_W$. Their block diagonal sum is thus a legitimate transformation of the whole square map, proving (ii).

Conversely, the three scalar polars of $Q_* \perp H_+^{\perp r}$ are blockwise nondegenerate. Its original Gauss sums are $16 \cdot 4^r = 2^n$, so $s = N + 3h$. Its dual character sum is $-64 \cdot 16^r = -2^{2n-2}$. Thus $\mathcal{P} = -2^{5n-2}$, and (3.2) gives exactly $v_4 = M_n - 2^{3n-4}$. Strict equivalence preserves these quantities.

The group $H(K)$ can be written on $K^2 \times K$ with

$$((u, v), w)((u', v'), w') = ((u + u', v + v'),\ w + w' + uv').$$

It has square map uv, order 64, and center K. By lemma 2.2 and proposition 2.1, (ii) is equivalent to (iii). In the central product the kernel has order 4^r, and the order is

$$1024 \cdot 64^r / 4^r = 2^{10+4r} = 2^{2n+2}.$$

This proves all directions and the uniqueness of the group type. It asserts no uniqueness of a chosen core subspace. □

Remark 6.4 (Target changes and mixed input changes). The preceding proof already normalized both restrictions with the same target fixed. Center identifications do not create extra types here: every member of $\mathrm{GL}_{\mathbb{F}_2}(K)$ is $T(w) = cw^\sigma$, with $c \in K^\times$ and $\sigma = \mathrm{id}$ or the Frobenius square. These are six distinct maps, all the elements of $\mathrm{GL}_2(\mathbb{F}_2)$. Each has a lift on a positive plane,

$$(u, v) \longmapsto (cu^\sigma, v^\sigma), \qquad H_+(cu^\sigma, v^\sigma) = T(H_+(u, v)).$$

Thus a different center gluing on each positive factor can be absorbed by its input change, without imposing an incompatible target change on the core.

The remainder in lemma 6.3 need not consist of positive planes in an initially chosen splitting. For $A_-(u, v) = uv + u^2 + \alpha v^2$, each scalar original sign is negative, but $A_- \perp A_-$ is isometric to $H_+ \perp H_+$ by the same lemma. For example, a K-linear change with columns

$$(0, 1, 0, 1), \quad (0, 1, 1, 1), \quad (1, 0, 1, 0), \quad (\alpha, 1, \alpha, 0)$$

pulls the former back to the latter. Permitting such mixed input changes is necessary; initial plane signs are not invariants of the whole remainder. In particular $Q_* \perp A_- \perp A_-$ also attains the corresponding lower bound, by this target-fixed isometry and theorem 1.2.

7 Two distinct limitations of stronger conclusions

Example 7.1 (Maximality does not pass to orthogonal blocks)**.** On K^4 and K^2, with the same target K, take

$$Q_0^- = ad + bc + \mathrm{Tr}(ac) + \alpha a^2 + d^2, \qquad A_-(u, v) = uv + u^2 + \alpha v^2.$$

The three scalar polars of each are nondegenerate. In the core parameters these are $g = \alpha,\ j = 1$, with $f = \ell = m = \eta = k = \theta = 0$. Formula (5.5) gives each original core sign negative, and lemma 6.1 gives each original plane sign negative. Their Gauss sums are therefore respectively -16 and -4, while the product $Q_0^- \perp A_-$ has three original sums $+64$. The two restrictions have $s = 208$ and $s = 4$, although the whole group of order 16384 has the maximum $s = 4288$.

Here $D_0 = \mathrm{Tr}(\sigma v) + 1$ and $D_1 = 1$. The global dual character sum is $+1024$, not -1024. Since $M_6 = 803403$, equation (3.2) gives

$$v_4(Q_0^- \perp A_-) = 819787, \qquad v_4(Q_* \perp H_+) = 787019.$$

Thus this is a counterexample to blockwise positivity inferred from maximum s alone, not to theorem 1.2. The negative lower-equality sign in lemma 6.2 cannot be omitted.

Lemma 7.2 (The polar obstruction survives target changes)**.** *For a nondegenerate scalar pencil, a target basis change* $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2(\mathbb{F}_2)$ *changes* $A = B_1^{-1}B_2$*, up to input similarity, to*

$$(aI + bA)^{-1}(cI + dA). \tag{7.1}$$

The operator is semisimple before this change if and only if it is semisimple afterwards.

Proof. The new polar pair is $B_1(aI + bA), B_1(cI + dA)$. Its first member is nondegenerate by hypothesis, proving both the formula and the invertibility of the denominator. A semisimple operator diagonalizes over an algebraic closure, so any defined rational function of it is semisimple. Applying the inverse target transformation proves the converse. Input similarity also preserves semisimplicity. □

Example 7.3 (An odd-half-dimension lower equality without the core)**.** Let $L = \mathbb{F}_8 = \mathbb{F}_2[t]/(t^3 + t + 1)$, $d = t^2 + t + 1$. In $V = L^2 \oplus K^2$, with target $W = \mathbb{F}_2^2$, take

$$\begin{aligned} Q_{\mathrm{odd}}(x, y, u, v) = (&\,\mathrm{Tr}_8(xy) + \mathrm{Tr}(uv), \\ &\mathrm{Tr}_8(txy + x + dy) + \mathrm{Tr}(\alpha uv)). \end{aligned}$$

By lemmas 4.1 and 4.2, its scalar polars are nondegenerate, its original sums are $8 \cdot 4 = 32$, and its dual character sum is $-8 \cdot 16 = -128$. Thus $n = 5$, the group order is 4096, and

$$s = 1120, \qquad v_4 = M_5 - 2^{10} = 57467 - 1024 = 56443.$$

It attains the odd lower endpoint in theorem 1.1.

Its A has a pair of simple $t^3 + t + 1$-blocks and a pair of simple p-blocks, and is semisimple: under the trace pairing on $\mathbb{F}_8^2$, the two scalar polars correspond to multiplication by 1 and t, so A is multiplication by t on both coordinates; on K^2 it is multiplication by α. The core Q_* has a pair of p^2-blocks and is not semisimple. If Q_{odd} had a common orthogonal factor strictly equivalent to Q_*, the corresponding restriction of its A would be nonsemisimple after a common target change. This contradicts lemma 7.2. It therefore has no such factor, even with an unrestricted remaining summand. The same polar obstruction applies to the odd equality types in proposition 3.5(a); it does not classify their complete square maps.

8 Connections with existing results

The quadratic-map and central-product correspondences used in proposition 2.1 and lemma 2.2 are standard. For the latter the precise source is (Kaur and Kulshrestha, 2012, Lemma 2.4, pp. 6–7), numbered according to its arXiv v1 PDF. No strongly real hypothesis is required there.

The maximum-s branch is exactly the totally orthogonal branch within (1.1). There are N linear characters with Frobenius–Schur indicator $+1$, and three nonlinear characters χ_λ of degree h, inflated from the three extraspecial quotients (Kaur and Kulshrestha, 2015, Proposition 3.5 and Remark 3.6, p. 8). Their squared degrees sum to $N + 3h^2 = |G|$, proving exhaustiveness. The indicator definition yields

$$\nu(\chi_\lambda) = \frac{1}{4N} \sum_{g \in G} \chi_\lambda(g^2) = \frac{\mathcal{G}(\lambda Q)}{h}.$$

Thus all indicators are $+1$ exactly when $s = N + 3h$. This is the specialization of the Arf criterion restated in (Kaur and Kulshrestha, 2012, Theorem 4.1, p. 12), where it is attributed to Zahinda's Theorem 3.5. We use that attribution through the cited source, not as a claim to have checked every result of the original paper.

For binary output dimension two, the three actual scalar duals are distinct and nonzero: equality of two would contradict Fourier inversion, and the dual of a bent function is bent. Their set, with zero adjoined, is a two-dimensional function space if and only if their sum D is identically zero. Consequently $D \equiv 0$ is exactly vectorial dual-bent closure in (Çeşmelioğlu et al., 2018, Definition 1, p. 309) in this situation. Any relabeling of the three nonzero components comes from $\mathrm{GL}_2(\mathbb{F}_2)$. The matrix identity

$$B_1^{-1} + B_2^{-1} = B_1^{-1}(B_1 + B_2)B_2^{-1}$$

makes the sum of two quadratic duals bent, but does not identify it with the third dual.

On the maximum-s branch, $D \equiv 0$ supplies the same-label vectorial dual and satisfies Condition A of (Wang et al., 2023, p. 9): take $p = 2$, input dimension $2n$, output dimension $2 \le n$, regularity coefficient $\epsilon = 1$, and $F = Q$, with $F(0) = 0$. For $I = \{0\}$, (Wang et al., 2023, Theorem 6, equation (35), pp. 23–24) gives a partial difference set $E = Q^{-1}(0) \setminus \{0\}$ with

$$z = \frac{N + 3h}{4}, \qquad k = z - 1, \qquad \lambda = \frac{N}{16} + \frac{3h}{4} - 2.$$

If L counts the totally zero binary planes, the difference-set parameter counts their six ordered bases, giving $6L = k\lambda$. Therefore the elementary lift count in lemma 2.3 gives

$$v_4 = 16L + 6z - 5 = M_n + 2^{3n-2}.$$

The $D \equiv 0$ upper endpoint is thus a direct corollary. Binary regularity does not assert a positive zero-frequency Gauss sum on every orthogonal restriction. Neither $C = 0$ nor $\mathcal{G}(D) = 0$ can replace the condition $D \equiv 0$.

The lower-equality maps have $\operatorname{rank} C = 4$, so they are not themselves vectorial dual-bent. The completed-class result of (Çeşmelioğlu et al., 2018, Theorem 1, p. 311) concerns sets of scalar functions, not a common strict equivalence of their simultaneous refinements. Even allowing the full extended affine action $\widetilde{Q}(x) = TQ(Sx + a) + Lx + b$, the quadratic polars transform as $\widetilde{B}_\lambda = S^{\mathsf{T}} B_{T^*\lambda} S$, whence $\widetilde{C} = S^{-1} C S^{-\mathsf{T}}$. Translation and linear terms do not change these polars. Thus rank four also prevents a simultaneous extended-affine representative from becoming a closed rank-zero source. This does not deny separate membership of scalar components in a completed class. The group equivalence throughout remains (1.4), which is more restrictive.

The polar decompositions are instances of alternating-pencil theory. (Lewis and Maglione, 2020, Theorem 1.2) enumerates isoclinism classes; its Theorems 3.7–3.8 also impose exponent p. We do not use them as isomorphism classifications for the exponent-four groups here. In characteristic two, homogeneous binary quadratic polynomials can represent linear functions by diagonal squares. The local and mixed-length framework in (Plût et al., 2014, Section 4, pp. 15–18) therefore retains the full refinements. The equality module here is $R^2 \oplus (R/(\varepsilon))^{2r}$; its transformation formulas and the classical Arf criterion underlie the short local normalizations. Our explicit normalization does not depend on Proposition 7 of the saved nineteen-page version of (Macario-Rat et al., 2013). A counterexample to that particular printed statement is given in appendix B; no conclusion about the whole algorithm or other versions is inferred from it.

A The common Maiorana–McFarland coordinates

This appendix gives the parameter correspondence behind lemma 5.2. Let

$$E = K[\beta], \qquad \beta^2 + \beta = \alpha, \qquad \mathrm{Tr}_{E/K}(\beta) = 1.$$

Thus $E = \mathbb{F}_{16}$. Put

$$X = (b + d) + \beta d, \quad Y = a + \beta c, \qquad U = \upsilon + \beta\sigma, \quad V = (\tau + \rho) + \beta\rho.$$

Direct multiplication gives

$$\mathrm{Tr}_{E/K}(XY) = ad + bc, \qquad \mathrm{Tr}_{E/\mathbb{F}_2}(UX + VY) = \mathrm{Tr}(\rho a + \sigma b + \tau c + \upsilon d).$$

Write $\mathcal{H}(Y) = \mathrm{Tr}(ac) + fa + ga^2 + \ell c + mc^2$, interpreted as a K-valued function of $Y = a + \beta c$. For $z \in K^\times$, define

$$h_z = (z\theta + z^2 j^2) + \beta(z\eta + z^2 k^2), \quad \pi_z(Y) = zY + h_z, \qquad g_z(Y) = z\beta\mathcal{H}(Y).$$

The trace identity $\mathrm{Tr}(zw^2) = \mathrm{Tr}(z^2 w)$ then gives exactly

$$\mathrm{Tr}(zQ^{(0)}) = \mathrm{Tr}_{E/\mathbb{F}_2}\big(X\pi_z(Y) + g_z(Y)\big).$$

Each π_z is an affine permutation of E. Apply (Çeşmelioğlu et al., 2018, equation (4), p. 313) with $p = 2$, $m = 4$, $s = 1$, component parameter 1, and the trace pairing displayed above. The actual phase is

$$\mathrm{Tr}_{E/\mathbb{F}_2}\big(V\pi_z^{-1}(U) + g_z(\pi_z^{-1}(U))\big).$$

Here $\pi_z^{-1}(U) = a_z + \beta c_z$, with a_z, c_z from (5.4). Substitution gives (5.5) including $E_z(0)$, for every one of the 2^{16} refinements, not only the 96 selected maps. When $\eta = \theta = j = k = 0$, the whole K-valued core has the common representation $\mathrm{Tr}_{E/K}(XY + \beta\mathcal{H}(Y))$, with a generally nonzero arbitrary function g. The later $g = 0$ hypotheses of (Çeşmelioğlu et al., 2018, Theorem 2) are not needed for equation (4).

B A version-specific vanishing-pencil example

We refer only to the nineteen-page ePrint revision of (Macario-Rat et al., 2013), Proposition 7 on p. 14. The consulted PDF has SHA-256 `6ce819d937f24089cf0e81f508129808e681346464a1c75cb5642981ef62aa1d`. Its statement sends any vanishing pencil to characteristics $(0, 0, H^m, 0)$, $m \geq d/2$. The following example checks the claim at the smallest

nonzero nilpotent parameter. It does not belong to the semi-extraspecial class of section 1 and plays no role in the proofs of the main results.

Take base and residue fields $\mathbb{F}_2$, local length $d = 2$, $R = \mathbb{F}_2[H]/(H^2)$, $t = 1 + H$, and $\lambda(a + Hb) = b$. In coordinates $x = a + Hb$, $y = c + Hd$, set

$$q_\infty = ad + bc, \qquad q_0 = ad + bc + ac + a^2 + c^2.$$

Both polars are nondegenerate. In the block order $(b, a; d, c)$, the multiplication matrix and trace matrix are

$$M = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad TM = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

The minimal and characteristic polynomials of M are both $(X + 1)^2$, so this is a cyclic regular pencil of the required primary type. The first form has no diagonal terms, and the second has the two diagonal characteristics H, H: the full tuple is $(0, 0, H, H)$. Moreover the invertible binary change

$$(a, b, c, d) \longmapsto (a, b + c, c, d + a + c)$$

interchanges the two quadratic forms. Indeed its pullback sends q_∞ to q_0, and adding $ac + a^2 + c^2$ sends the latter back to q_∞. Thus both ordered pencils are semi-vanishing, which is exactly the printed vanishing condition.

For the input pair, simultaneous vanishing requires $ac + a + c = 0$, whose only solution in $\mathbb{F}_2^2$ is $a = c = 0$. There are therefore four common zeros. The printed targets at $d = 2$ have second form either $ad + bc + ac + a^2$ or $ad + bc + ac$. In the first case the permitted (a, c) are $(0, 0), (0, 1), (1, 1)$; in the second they are $(0, 0), (0, 1), (1, 0)$. Each gives respectively four, two, and two choices of (b, d), so each target has eight common zeros. An invertible input change preserves this number, as does even a common invertible target change. Hence the printed reduction is false for this example over the original field. No field extension or affine operation is part of that fixed-field isomorphism statement. This observation is restricted to that version and proposition; it does not invalidate its other local transformation formulas or claim anything about originality of the main results here.

Use of generative AI

Generative AI tools, including OpenAI Codex/ChatGPT and the system recorded as Muse Spark 1.3, contributed to mathematical exploration and proof development, computational checks, literature review, and manuscript drafting. This included substantive mathematical steps and initial text generation, not only language editing. Exact historical deployment versions are not fully documented. Internal automated checks are not external peer review.

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